

The Baxterised Ω is an Isaev–Kirillov Hecke R -factor staircase: an exact spectral bridge

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Abstract

Clio's order law $\text{ord}_{x=q^2} Z_\lambda(x) = \tau(\tau + 1)/2$ governs a staircase monodromy $M(x)$ of Baxterised Hecke R -factors evaluated at a degenerate spectral point $x^* = q^2$. We prove that each such R -factor is, under an explicit Möbius reparametrization of the spectral variable, *exactly* the normalized Baxterised Hecke generator of Isaev–Kirillov (arXiv:1302.6495), whose Markov-traced staircases generate the Bethe commutative subalgebra of $H_q(S_n)$. Consequently the trace $Z_\lambda(x) = \text{tr}_{V^\lambda} M(x)$ belongs to the Bethe-subalgebra trace family, and the order law is the vanishing order of a Bethe transfer-matrix trace at a fusion point. We record the bridge, its verification, and an honest account of what remains to identify Ω as a specific Bethe-subalgebra coefficient.

1 Setup: Clio's object

In the Hecke algebra $H_q(S_n)$ with generators T_i satisfying

$$(T_i - q)(T_i + q^{-1}) = 0,$$

let

$$P_q^{(i)} = \frac{T_i + 1}{q + 1}, \quad P_{-1}^{(i)} = \frac{qI - T_i}{q + 1}$$

be the projectors onto the $T_i = q$ and $T_i = -q^{-1}$ eigenspaces. (One checks $P_q^{(i)} + P_{-1}^{(i)} = \frac{(T_i + 1) + (qI - T_i)}{q + 1} = \frac{(q + 1)I}{q + 1} = I$; these are the idempotents I use throughout.) The Baxterised factor is

$$\check{R}_i(x) = P_q^{(i)} + r(x) P_{-1}^{(i)}, \quad r(x) = \frac{q^2 - x}{q(x - 1)}.$$

The monodromy $M(x) = \prod_k \check{R}_{i_k}(x)$ is the ordered product over a staircase reduced word; $Z_\lambda(x) = \text{tr}_{V^\lambda} M(x, \dots, x)$ and $\Omega = (q + 1)^N M(q^2)$.

The special values of r organize the geometry of the spectral line:

- $r(q) = 1$, so $\check{R}(q) = I$: the *regular* point;
- $r(q^2) = 0$, so $\check{R}(q^2) = P_q$: the *fusion* point onto the symmetric line;
- r has a pole at $x = 1$: the other fusion, onto P_{-1} .

The order law, proved for hooks and intact-junction off-hooks, reads

$$\text{ord}_{x=q^2} Z_\lambda = \frac{\tau(\tau + 1)}{2}, \quad \tau = \max(0, 2\lambda'_1 - n - 1).$$

The whole edifice is built from a single one-parameter family of operators $\check{R}_i(x)$ on the Hecke algebra. It has long bothered me that this family looks like it ought to be *somebody's* Baxterised generator. This note shows it is.

2 The Isaev–Kirillov Baxterised generator and Bethe subalgebra

Isaev–Kirillov (arXiv:1302.6495, Lett. Math. Phys. **104** (2014) 179) work on the same Hecke algebra. They write the quadratic relation as $T_i^2 = 1 + \lambda T_i$ with $\lambda := q - q^{-1}$, which is the same relation $(T_i - q)(T_i + q^{-1}) = 0$. They define the Baxterised generator

$$T_i(x) = (1 - x)T_i + \lambda x,$$

with normalized form

$$\tilde{T}_i(x) = \frac{T_i(x)}{q - q^{-1}x}$$

satisfying unitarity $\tilde{T}_i(x)\tilde{T}_i(x^{-1}) = I$ and the multiplicative-spectral Yang–Baxter equation.

They then build the Baxterised Jucys–Murphy element as a sandwiched staircase product

$$y_n(x; \vec{\xi}) = T_{n-1}(x/\xi_{n-1}) \cdots T_1(x/\xi_1) y_1(x) T_1(x\xi_1) \cdots T_{n-1}(x\xi_{n-1}),$$

and the Bethe commutative subalgebra $\mathcal{B}_n(\vec{\xi}) \subset H_q(S_n)$ is generated by the coefficients Φ_k of the Markov trace

$$\tau_n(x) = \text{Tr}_{(n+1)}(y_{n+1}(x; \vec{\xi})) = \sum_k \Phi_k x^k.$$

Commutativity $[\tau_n(x), \tau_n(z)] = 0$ is their Proposition 3.2. In the $q \rightarrow 1$ limit the Φ_k degenerate to the Gaudin Hamiltonians.

One bookkeeping difference must be flagged before any comparison. The IK spectral parameter is *multiplicative*: their Yang–Baxter equation involves the product xz , not the sum $x + z$. My $r(x)$ is written *additively*. This convention gap is exactly what the Möbius bridge of the next section closes.

3 Main Lemma: the spectral conjugacy

All identities in this section were verified symbolically over $\mathbb{Q}(q)$ in sympy.

Lemma 1 (Eigenvalues). *On the $T_i = q$ eigenline, $T_i(x)$ acts by $q - q^{-1}x$; on the $T_i = -q^{-1}$ eigenline, by $qx - q^{-1}$. Hence the normalized generator decomposes as $\tilde{T}_i(x) = P_q + \rho(x)P_{-1}$ with eigenvalue-ratio*

$$\rho(x) = \frac{qx - q^{-1}}{q - q^{-1}x} = \frac{q^2x - 1}{q^2 - x}.$$

Proof. On the $T_i = q$ eigenline, $T_i(x) = (1 - x)q + \lambda x = q - qx + (q - q^{-1})x = q - q^{-1}x$. On the $T_i = -q^{-1}$ eigenline, $T_i(x) = (1 - x)(-q^{-1}) + \lambda x = -q^{-1} + q^{-1}x + (q - q^{-1})x = -q^{-1} + qx = qx - q^{-1}$. Dividing each by the normalization $q - q^{-1}x$ gives $\tilde{T}_i(x)$ acting by 1 on the $T_i = q$ eigenline and by $\rho(x) = (qx - q^{-1})/(q - q^{-1}x)$ on the $T_i = -q^{-1}$ eigenline. Multiplying numerator and denominator by q yields $\rho(x) = (q^2x - 1)/(q^2 - x)$. $\square$

Theorem 1 (Möbius spectral bridge). *Define the Möbius reparametrization*

$$\varphi(x) = \frac{q((q^3 - 1) - (q - 1)x)}{(q^3 - 1)x - q^2(q - 1)}.$$

Then

$$\rho(\varphi(x)) = r(x) = \frac{q^2 - x}{q(x - 1)}$$

identically as rational functions in q, x . Equivalently, since both $\tilde{R}_i$ and $\tilde{T}_i$ have P_q -coefficient equal to 1,

$$\boxed{\tilde{R}_i(x) = \tilde{T}_i(\varphi(x))} \quad (\text{exact equality of operators}).$$

Proof. Both operators are $P_q + (\text{scalar}) \cdot P_{-1}$ in the same idempotent decomposition $I = P_q^{(i)} + P_{-1}^{(i)}$. The P_q -coefficients agree (both equal 1). The P_{-1} -coefficients are $r(x)$ on the left and $\rho(\varphi(x))$ on the right; their equality is the displayed rational-function identity, verified symbolically over $\mathbb{Q}(q)$. Equality of the two scalar coefficients forces equality of the operators. $\square$

Corollary 1 (Special points). $\varphi(q) = 1$, $\varphi(q^2) = q^{-2}$, and $\lim_{x \rightarrow 1} \varphi(x) = q^2$. Thus the dictionary of distinguished points is:

- *regular* ($\check{R} = I$): *Clio* $x = q \leftrightarrow$ *IK* $x = 1$;
- *fusion* ($\check{R} = P_q$): *Clio* $x = q^2 \leftrightarrow$ *IK* projector point $x = q^{-2}$;
- *pole* ($\check{R} = P_{-1}$): *Clio* $x = 1 \leftrightarrow$ *IK* $x = q^2$.

Remark 1 (Normalization invariance of the order). If one uses the unnormalized IK $T_i(x)$ instead of $\tilde{T}_i$, each factor is rescaled by the scalar

$$f(x) = q - q^{-1}\varphi(x) = \frac{(q+1)(q^2+1)(x-1)}{q^2x - q^2 + qx + x}.$$

At the fusion point $f(q^2) = q - q^{-3}$, which is finite and nonzero. Hence the choice of normalized versus unnormalized IK generator multiplies $M(x)$ near $x = q^2$ by a finite nonzero scalar and does *not* change $\text{ord}_{x=q^2} Z_\lambda$. (All four identities above were verified symbolically over $\mathbb{Q}(q)$.)

4 Consequence: Ω lives in the Bethe construction

Because $\check{R}_i(x) = \tilde{T}_i(\varphi(x))$ factor by factor, Clio's staircase $M(x)$ is, after the single global reparametrization $x \mapsto \varphi(x)$, an ordered staircase of genuine IK normalized Baxterised Hecke generators — the same architecture as the IK Baxterised Jucys–Murphy element y_n . Therefore:

- $Z_\lambda(x) = \text{tr}_{V^\lambda} M(x)$ is a trace of a reparametrized IK transfer-type staircase: it lives in the Bethe-subalgebra *trace* family.
- Clio's degenerate evaluation $x^* = q^2$ is, under φ , the IK projector/fusion point $x = q^{-2}$ where the normalized generator collapses to P_q (the symmetric projector). So Ω is the fusion-point value of a traced staircase of IK Baxterised generators.
- Hence the order law $\text{ord}_{x=q^2} Z_\lambda = \tau(\tau+1)/2$ is the order of vanishing of a Bethe transfer-matrix trace at a fusion point — a q -KZ degeneration order in the Isaev–Kirillov–Tarasov (arXiv:1510.05374) sense, where these Baxterised elements furnish flat connections for the q -KZ equations.
- Clio's deficit law (deleting u tail factors drops the E^- -sign-kill depth by exactly u) is then the concrete Hecke-basis computation of this degeneration's ramification: *delete-u-drop-u* is *merge-one-fewer-mode*.

The pleasing thing is that nothing in this consequence is a new conjecture. Once the per-factor identity is exact, the architecture transports verbatim; what changes is only the interpretive frame in which Ω now sits.

5 Honest scope and open questions

What is proven here. The per-factor identity $\check{R}_i(x) = \tilde{T}_i(\varphi(x))$ and the special-point dictionary, exactly and symbolically. This places $M(x)$ and Z_λ in the IK Baxterised-staircase / Bethe-trace family and identifies $x = q^2$ as the fusion point.

What is not yet established.

- The IK Bethe subalgebra is generated by the *Markov* trace of y_{n+1} , whereas Z_λ is a V^λ -character (an irreducible-module trace). Matching these requires expressing the V^λ -trace via Markov-trace coefficients — a Geck–Rouquier / cocenter step.
- The IK seed $y_1(x)$ and the inhomogeneities $\vec{\xi}$ have not been matched to Clio's homogeneous evaluation $\vec{\xi} = (1, \dots, 1)$, $x \rightarrow x$.

- (c) The quantitative count $\tau(\tau + 1)/2$ is pinned to Clio's normalization; only the *order* (not the leading coefficient) is shown normalization-invariant.

Until (a)–(b) are settled, the correct statement is: Ω is structurally an IK Baxterised-staircase element evaluated at the fusion point, and a Bethe-subalgebra trace up to the trace-type identification.

Next step: the sharpened decisive test. Build the $(\tau + 1) \times (\tau + 1)$ confluent discrete Wronskian of the colliding antisymmetric-excess modes and verify that its ramification order is $\binom{\tau+1}{2}$ *and* reproduces the deficit law's delete- u -drop- u arithmetic. With the bridge in hand, this becomes an operator identity in $H_q(S_n)$ attackable with seminormal tools — precisely the kind of statement that the spectral conjugacy was meant to make legible.